Committing Perishable Supply Under Partial Information

A cost optimizer for operators who must allocate before they know what arrives

Prepared for Dr. Alex Psomas · Rad Al Zubi and Mustafa Sabieha, Industrial Engineering, Purdue University

WHY WE ARE WRITING TO YOU

Your work on dynamic fair division asks whether an allocator can still act well when it receives only partial information about the agents it serves, and Food Drop puts that question into a real food system. We are working on the same structural problem one step upstream. A midstream perishable operator, meaning a distributor, aggregator, or food hub, must commit supply to buyers weeks before learning what its own growers will actually deliver. The information arrives after the decision is irreversible. Our objective is not fairness across recipients but expected cost to a single operator, since a wrong commitment resolves either as wasted product or as a missed order.

WHAT THE SYSTEM DOES, END TO END

1. **Ingest and structure.** Messy CSV exports from spreadsheets or an ERP land in staging tables under hard validation. Negative quantities, reversed date pairs, and unparseable dates are rejected with named rows rather than silently coerced. A leakage guard asserts that every date in a training or evaluation slice falls inside its declared window, and refuses rather than passing when it cannot check.
2. **Diagnose.** Descriptive arithmetic on the operator's own records: waste, shortage, supplier fill and rejection, buyer fulfillment, and available to promise. Auditable rather than learned, and deliberately so, because this is the layer that earns trust first.
3. **Forecast demand.** A global gradient boosted quantile model with Croston's method routing intermittent series, quantile crossing repaired by monotone rearrangement (Chernozhukov, Fernandez-Val and Galichon), and intervals corrected by conformalized quantile regression (Romano, Patterson and Candes), which buys distribution-free finite-sample coverage. Where calibration data is too thin to support the guarantee, the calibrator refuses instead of reporting a false one.
4. **Forecast supply.** Supplier delivery reliability as an empirical Bayes Beta-Binomial, pooling thin-history suppliers toward the population in both mean and variance, so three near-identical deliveries cannot masquerade as certainty.
5. **Build joint scenarios.** Supplier dependence through Gaussian and t copulas under Sklar's theorem, with rank correlation mapped to the latent parameter by the elliptical Kendall identity and the estimated matrix regularized by Ledoit-Wolf style shrinkage. This mattered: independent draws understate aggregate risk enough to erase the value of stochastic modeling entirely.
6. **Optimize.** A two stage stochastic program with recourse (Birge and Louveaux) solved by sample average approximation. Commitments are first stage; spot purchases, waste, and unmet demand are second stage recourse. The objective is newsvendor rather than gross cost, so the solution tracks the critical fractile of the underage and overage ratio.
7. **Validate, then report.** Every run carries provenance: which dependence model was used, which seed, and whether any fallback was taken and why.

HOW WE CHECK OURSELVES, AND THE RESULT WE LIKE LEAST

Forecasts are scored with proper scoring rules, meaning pinball loss, the weighted interval score, and CRPS, with PIT histograms for calibration. The optimizer is judged by the value of the stochastic solution, computed out of sample with a bootstrap interval and reported as beneficial, harmful, or inconclusive rather than assumed positive. On a 52 week synthetic operator it beats naive and order-up-to policies everywhere tested, but against a deterministic linear program the advantage is contingent on spot economics: plus 11.1 percent when emergency coverage is cheap and plentiful, minus 4.0 percent when it is expensive and rationed. We surface that rather than bury it.

WHERE WE ARE, PLAINLY

Built and gate tested, but every number above comes from a calibrated synthetic operator. No real operator's records have touched any layer. No customers, and we are not raising money.

WHAT WE WOULD VALUE FROM A CONVERSATION

- **The question only you can answer.** We treat supplier capacity and yield reports as truthful. Suppliers competing for commitments have obvious incentives to misreport, and we currently have no strategic layer at all. Is that a fatal omission or a reasonable simplification at this stage?
- **Deployment and trust.** How did the Indy Hunger Network come to accept a system that makes allocations rather than recommendations? That transition is the one we understand least.
- **Framing and critique.** Is the pre-commitment decision genuinely where the loss originates, and where does this pipeline break?
- **People and data.** Anyone in the Food Drop orbit, a food bank, a grocery or restaurant donor, a logistics partner, who might recognize this problem or share historical records for a free, no obligation analysis. We sign an NDA first.

Rad Al Zubi and Mustafa Sabieha
Industrial Engineering, Purdue University